WORKED SOLUTION

STEP I 1999, Question 14

Probability and Statistics • Conditional Probability

STEP I 1999, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q14

The key idea

The continuous random variable is only there to set up a discrete one. Once we turn the density into the four probabilities of scoring 4, 3, 2 and 1, part (i) is a one-line expectation. Part (ii) is a conditional probability, and since the total must hit 3 exactly, there are very few ways to win, so we can list them all.

Solution

Part (i)

For $0 \le a \le b \le 1$,

$$\mathrm{P}(a \le X < b) = \int_a^b 2x \,\mathrm{d}x = b^2 - a^2.$$

(Including or excluding an endpoint makes no difference for a continuous random variable.) Let S be the score from one dart. Then

$$\mathrm{P}(S=4) = \left(\tfrac{1}{4}\right)^2 - 0 = \tfrac{1}{16}, \qquad \mathrm{P}(S=3) = \left(\tfrac{1}{2}\right)^2 - \left(\tfrac{1}{4}\right)^2 = \tfrac{3}{16},$$

$$\mathrm{P}(S=2) = \left(\tfrac{3}{4}\right)^2 - \left(\tfrac{1}{2}\right)^2 = \tfrac{5}{16}, \qquad \mathrm{P}(S=1) = 1 - \left(\tfrac{3}{4}\right)^2 = \tfrac{7}{16}.$$

These add up to 1, as they should. Notice that the outer rings are more likely: the density $2x$ grows with x, because there is more area far from the centre.

So

$$\mathrm{E}(S) = 4 \cdot \tfrac{1}{16} + 3 \cdot \tfrac{3}{16} + 2 \cdot \tfrac{5}{16} + 1 \cdot \tfrac{7}{16} = \frac{4+9+10+7}{16} = \frac{30}{16} = \frac{15}{8},$$

as required.

Part (ii)

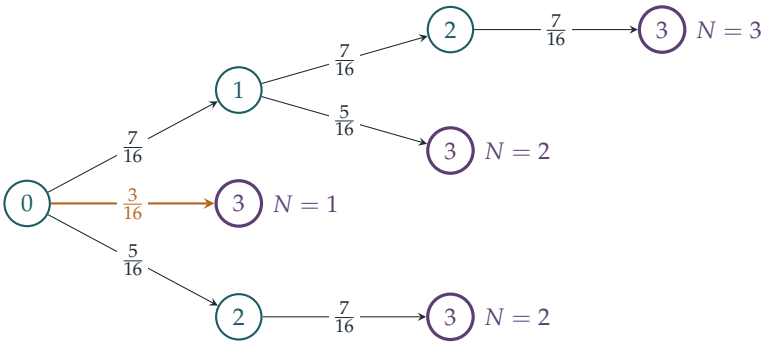
Assume the throws are independent, each with the distribution of S above. Let W be the event that I win, and N the number of throws in the game.

Ask yourself Which sequences of scores lead to a win?

I win exactly when the running total reaches 3 without first going past it. Every score is at least 1, so the total goes up by at least 1 each time and the game lasts at most three throws. A winning game is a sequence of positive scores that adds up to exactly 3. (I only throw again while the total is below 3, so every such sequence really is a complete game.) There are just four:

$$(3), \qquad (1,2), \qquad (2,1), \qquad (1,1,1).$$

A score of 4 never appears, because it takes the total past 3 at once.



The tree shows the running totals along each route that ends in a win (circled in purple), labelled with the probability of each throw.

By independence, the probability of each sequence is the product of the probabilities of its scores:

$$\begin{aligned}
\mathrm{P}(W \cap N = 1) &= \mathrm{P}(3) = \tfrac{3}{16} = \tfrac{768}{4096}, \\
\mathrm{P}(W \cap N = 2) &= \mathrm{P}(1,2) + \mathrm{P}(2,1) = 2 \cdot \tfrac{7}{16} \cdot \tfrac{5}{16} = \tfrac{70}{256} = \tfrac{1120}{4096}, \\
\mathrm{P}(W \cap N = 3) &= \mathrm{P}(1,1,1) = \left(\tfrac{7}{16}\right)^3 = \tfrac{343}{4096}.
\end{aligned}$$

These three events are disjoint and together make up W, so

$$\mathrm{P}(W) = \frac{768 + 1120 + 343}{4096} = \frac{2231}{4096}.$$

Therefore

$$\mathrm{P}(N = 3 \mid W) = \frac{\mathrm{P}(W \cap N = 3)}{\mathrm{P}(W)} = \frac{343/4096}{2231/4096} = \frac{343}{2231},$$

as required.

Checks

- The four score probabilities $\frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}$ add up to 1. They are odd numbers over 16 because $(k+1)^2 - k^2 = 2k + 1$.

- The target $343 = 7^3$ shows straight away that the three-throw win must be $(1,1,1)$, which confirms the setup.
- A computer simulation of 10^6 games gave $\mathrm{P}(W) \approx 0.5452$ (exact value 0.5447) and $\mathrm{P}(N = 3 \mid W) \approx 0.1535$ (exact value $343/2231 \approx 0.1537$).

Common slips

- Treating the score bands as equally likely. The probabilities come from integrating the density, not from the widths of the intervals.
- Counting $(1,2)$ but forgetting $(2,1)$, which gives $\mathrm{P}(W) = \frac{1671}{4096}$.
- Answering $\mathrm{P}(N = 3 \text{ and } W) = \frac{343}{4096}$ instead of dividing by $\mathrm{P}(W)$.

Beyond STEP

Part (ii) is a small Markov chain: the state is the running total, and 3 and "bust" are absorbing states. For bigger targets, listing the routes stops being practical. Instead we write w_t for the probability of winning from total t, which gives a recurrence $w_t = \sum_s \mathrm{P}(S = s)\, w_{t+s}$. This first-step analysis is a key tool in first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers it in their treatment of random walks and Markov chains.

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